

PSYCH-BENCH: A Comprehensive Cognitive Battery for Evaluating Holographic Liquid Brain Architectures

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Abstract

We present PSYCH-BENCH, a comprehensive psychological benchmark suite comprising 78 benchmarks across 17 cognitive domains, designed to evaluate Holographic Liquid Brain (HLB) architectures—neuroscience-inspired systems combining Hyperdimensional Computing (HDC), Liquid Time-Constant networks (LTC/CfC), Integrated Information Theory (IIT), and Active Inference. Unlike existing AI evaluation frameworks that either lack psychometric rigor (HELM), cover narrow domains (ARC), omit normative human comparison (BIG-Bench), or ignore neuromodulatory effects (CogBench), PSYCH-BENCH simultaneously addresses all four gaps. The suite provides: (1) psychometrically validated implementations of established paradigms (Stroop, WCST, N-back, Iowa Gambling Task, and 74 others) with exact procedural fidelity to source publications; (2) normative anchoring against 90+ published human baselines enabling z-score comparison and clinical-band interpretation; (3) a 17-benchmark neuromodulator evaluation framework testing dopaminergic reward learning, noradrenergic arousal curves, serotonergic mood induction, cholinergic attention, pharmacokinetic injection challenges, dose-response curves, tolerance/withdrawal dynamics, and multi-transmitter synergy; and (4) a 14-indicator consciousness evaluation matrix derived from six theories of consciousness. All stimuli are encoded through a unified holographic representation via five HDC adapter types, enabling cross-modal comparison. The harness provides ICC(3,1) test-retest reliability, ex-Gaussian reaction-time decomposition, speed-accuracy tradeoff curves, adaptive staircase procedures, and self-contained HTML reporting. Implemented in 15,000 lines of Rust with 793 unit tests, PSYCH-BENCH is open-source and fully deterministic given fixed seeds. We demonstrate the suite on the SYMTHAEA HLB, producing an 8-domain cognitive profile that reveals how holographic encoding drives pattern-matching strengths while liquid temporal dynamics govern attention and motor learning.

Keywords: cognitive benchmark, holographic computing, liquid neural networks, neuromodulation, psychometrics, consciousness evaluation, hyperdimensional computing

1 Introduction

A new class of cognitive architecture has emerged at the intersection of hyperdimensional computing, dynamical systems neuroscience, and consciousness science. These *Holographic Liquid Brain* (HLB) architectures encode information as high-dimensional distributed representations—the “holographic” component, drawing on Plate’s holographic reduced representations [50] and Kanerva’s hyperdimensional computing framework [32]—while processing temporal dynamics through liquid-state computation, specifically Liquid Time-Constant (LTC) networks [28] and their closed-form continuous-time variant (CfC) [29]. The “liquid” component provides biologically plausible temporal processing with continuous-time dynamics and adaptive time constants, while integration with Integrated Information Theory (IIT) [47, 61] and Active Inference under the Free Energy Principle [23, 24] creates architectures with rich internal dynamics that demand correspondingly rich evaluation.

The fundamental challenge is methodological: how should we evaluate cognitive architectures that aspire to neuroscience-grade fidelity? Standard AI benchmarks fall short in four critical ways.

Lack of psychometric rigor. The Holistic Evaluation of Language Models (HELM) [38] evaluates across many tasks but treats each as an isolated accuracy measure without psychometric validation—no test-retest reliability, no construct validity analysis, no speed-accuracy decomposition. A benchmark that fluctuates wildly across runs tells us nothing stable about the architecture it evaluates.

Narrow domain coverage. The Abstraction and Reasoning Corpus (ARC) [9] provides an excellent measure of fluid intelligence but covers a single cognitive domain. Human cognition is characterized by the interplay between executive function, working memory, social cognition, motor learning, and dozens of other capacities. Single-domain benchmarks cannot capture the profile that emerges from an integrated architecture.

Absence of normative comparison. BIG-Bench [57] offers hundreds of tasks but provides no systematic comparison to human performance distributions. Without normative anchoring—z-scores against published baselines with known means and standard deviations—we cannot interpret whether an architecture’s performance on a given task is exceptional, average, or impaired relative to the human population it models.

The neuromodulator gap. Perhaps most critically, no existing benchmark suite tests the effects of pharmacological manipulations on cognitive performance. CogBench [12] applies eight cognitive psychology paradigms to large language models, representing the closest existing work, but its eight tasks and prompt-based interface cannot probe neuromodulatory dynamics. For HLB architectures with explicit dopaminergic, noradrenergic, serotonergic, and cholinergic subsystems, evaluating how

these transmitters modulate behavior is essential—analogous to how pharmacological challenge studies validate computational models of the basal ganglia [18] or prefrontal cortex [1].

PSYCH-BENCH addresses all four gaps simultaneously. We present a comprehensive cognitive battery of 78 benchmarks spanning 17 domains, designed specifically for evaluating HLB and related neuroscience-inspired architectures. The suite is organized around five design principles: psychometric fidelity to source paradigms, unified holographic stimulus encoding, normative anchoring against published baselines, systematic ablation transparency, and self-contained reporting.

Our contributions are:

1. A 78-benchmark cognitive battery covering 8 profile domains (executive function, memory, attention, social cognition, motor learning, language, metacognition, reasoning) plus specialized sub-batteries for affect, creativity, inhibition, and consciousness.
2. A 17-benchmark neuromodulator evaluation framework—the first to test pharmacological manipulations (dose-response curves, injection challenges, tolerance/withdrawal, antagonist profiles, multi-transmitter synergy) in an AI benchmark context.
3. A psychometric infrastructure providing ICC(3,1) reliability, ex-Gaussian RT decomposition [30], Wickelgren SAT curves [63], adaptive staircases [37], and normative z-score comparison with clinical interpretation bands [10].
4. A 14-indicator consciousness evaluation derived from Butlin et al.’s [8] six-theory framework, with ablation matrices linking architectural features to indicator presence.

2 Design Principles

PSYCH-BENCH is organized around five principles that distinguish it from existing AI evaluation suites.

Psychometric fidelity. Each benchmark implements the exact experimental paradigm described in its source publication. The Stroop task [60] presents congruent and incongruent color-word stimuli following MacLeod’s [41] standardized procedure. The Wisconsin Card Sorting Test [6] implements 6-category set-shifting with perseverative error tracking. The Iowa Gambling Task [5] uses the original reward/punishment schedule across 100 trials with four decks. This is not approximation—each benchmark carries a citation to its source paradigm, and the implementation can be audited against the original procedure.

Holographic stimulus encoding. All stimuli pass through a unified Hyperdimensional Computing (HDC) encoding pipeline before reaching the cognitive architecture. Five adapter types—SequenceAdapter, SemanticAdapter, SpatialAdapter, ScenarioAdapter, and RewardAdapter—convert raw stimuli (token sequences, spatial coordinates, reward outcomes, scenarios) into continuous hypervectors in $\mathbb{R}^D$ using binding ($\otimes$) and bundling ($\oplus$) operations [34]. This unified representation is the “holographic” in HLB: all modalities share a common geometric space where similarity is

cosine distance, enabling cross-modal retrieval and comparison. The default dimension $D = 512$ balances representational capacity against computational cost for benchmark-scale evaluation.

Normative anchoring. Every benchmark has at least one associated human baseline from the published literature, stored as a mean $\pm$ SD pair with citation. The `BaselineCollection` contains 90+ such entries spanning all 17 domains. Agent performance is converted to z-scores ($z = (x_{\text{agent}} - \mu_{\text{human}})/\sigma_{\text{human}}$, negated for error metrics where lower is better) and mapped to percentiles via the normal cumulative distribution function. Clinical interpretation bands follow Cicchetti’s [10] guidelines: Superior ($z > 2.0$), Above Average ($1.0 < z \leq 2.0$), Average ($-1.0 \leq z \leq 1.0$), Below Average ($-2.0 \leq z < -1.0$), and Impaired ($z < -2.0$).

Ablation transparency. Six predefined ablation presets systematically disable architectural subsystems: `FullConsciousness` (all systems active), `CfcOnly` (liquid dynamics only, no active inference or social cognition), `NoFep` (Free Energy Principle disabled), `NoSocial` (Theory of Mind disabled), `ReducedWm` (working memory capacity reduced from 7 to 3), and `HdcOnly` (minimal holographic encoding). Each preset produces a complete cognitive profile, enabling researchers to identify which subsystems drive performance on which tasks—a form of computational lesion study.

Self-contained reporting. The harness generates self-contained HTML reports with embedded SVG visualizations: radar charts for the 8-domain cognitive profile, forest plots of normative z-scores with 95% confidence intervals, speed-accuracy tradeoff curves, and reliability summaries. Reports require no external dependencies and can be archived alongside experimental results.

3 Architecture

PSYCH-BENCH consists of three layers (Figure 1): a benchmark layer defining experimental paradigms, a harness layer providing psychometric infrastructure, and a reporting layer generating analysis outputs.

3.1 Benchmark Trait and Configuration

All benchmarks implement the `PsychBenchmark` trait, which defines a uniform interface for stimulus presentation, response collection, and metric computation. The trait requires three methods: `name()` returning a domain-qualified identifier (e.g., `"Executive::Stroop"`), `run(config)` executing the full experimental procedure under a given `BenchmarkConfig`, and `expected_metrics()` listing the metric names and their directionality (higher-is-better or lower-is-better).

The `BenchmarkConfig` structure controls 17 parameters spanning four categories. *HDC parameters*: `dimension` (default 512), `ssm_backend` (state-space model alternative). *Experimental parameters*: `trials_per_condition` (default 20), `seed` (default 42), `time_pressure` (0.0–1.0 for

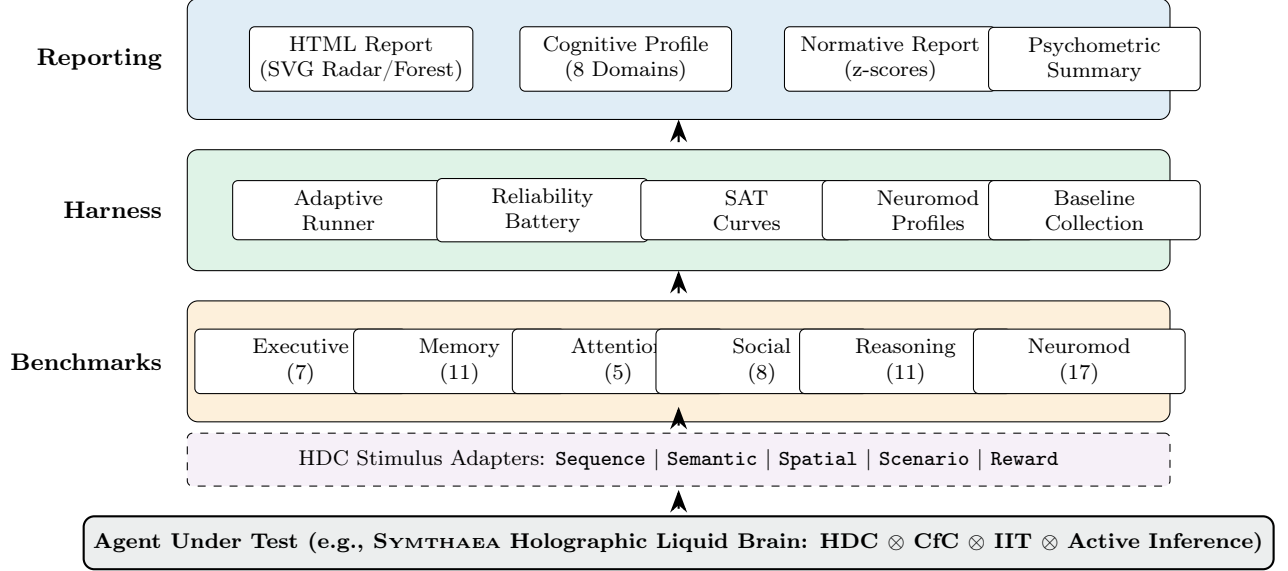


Figure 1: Three-layer architecture of PSYCH-BENCH. Benchmarks define experimental paradigms with HDC-encoded stimuli. The harness provides psychometric infrastructure (reliability, SAT curves, normative comparison, neuromodulator profiles). The reporting layer generates self-contained analysis outputs. Arrows indicate data flow from the agent under test upward through encoding, evaluation, and reporting.

SAT curves), `difficulty` (0.0–1.0 ramp). *Architecture parameters*: `working_memory_capacity` (default 7), `enable_social` (Theory of Mind), `enable_fep` (Active Inference), `planning_horizon` (FEP lookahead), `action_temperature` (softmax temperature). *Adaptive parameters*: `adaptive_trials`, `min_trials`, `max_trials`, `precision_target` (CI half-width / |mean|). A deterministic trial-level seed function ensures reproducibility: `trial_seed(benchmark, condition, trial)` produces a unique seed per trial while maintaining cross-run consistency.

3.2 Holographic Stimulus Encoding

The encoding pipeline (Figure 2) converts raw stimuli into continuous hypervectors via five adapter types, each implementing the `StimulusAdapter` trait.

The **SequenceAdapter** encodes ordered token sequences by generating a random basis hypervector per token and applying position-dependent permutations: $\mathbf{h}_{\text{seq}} = \bigoplus_i \text{basis}(t_i) \cdot P^i$. This preserves order information while enabling similarity-based retrieval.

The **SemanticAdapter** encodes words via FNV-1a hashing to generate basis vectors, with position-dependent binding to create compositional representations. For Remote Associates Test (RAT) stimuli, solution hypervectors are constructed as weighted bundles: $\mathbf{h}_{\text{sol}} = 0.79 \cdot \mathbf{h}_{\text{base}} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_0} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_1} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_2}$, ensuring that cue-solution associations are geometrically encoded.

The **SpatialAdapter** encodes 2D grid positions via conjunctive binding: $\mathbf{h}_{(x,y)} = \mathbf{h}_x \otimes \mathbf{h}_y^P$, where $\mathbf{h}_x$ and $\mathbf{h}_y$ are axis-specific basis vectors and P is a single permutation distinguishing dimensions. The companion **VisualObjectAdapter** extends this to feature conjunctions: $\mathbf{h}_{\text{obj}} = \mathbf{h}_{\text{color}} \otimes \mathbf{h}_{\text{shape}} \otimes \mathbf{h}_{\text{size}}$.

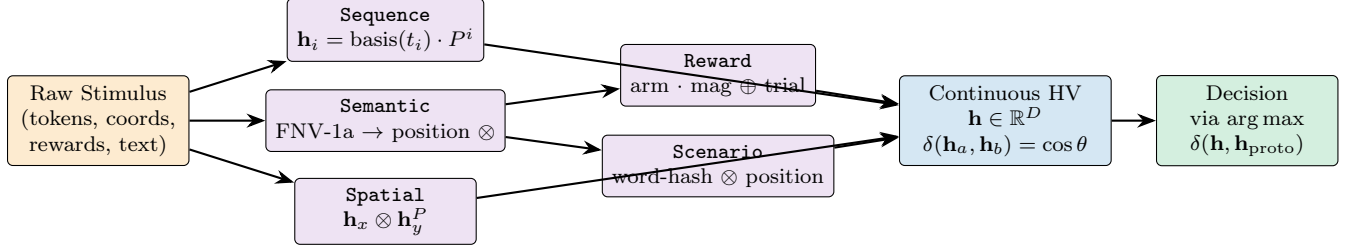


Figure 2: HDC stimulus encoding pipeline. Raw stimuli are converted to continuous hypervectors via five adapter types using binding ($\otimes$), bundling ($\oplus$), and permutation (P) operations. All modalities share a common $\mathbb{R}^D$ space where cosine similarity enables cross-modal comparison. Decisions use prototype matching via $\arg \max$ similarity.

The **RewardAdapter** encodes reinforcement learning outcomes by binding arm identity with reward magnitude and trial index: $\mathbf{h}_{\text{reward}} = \mathbf{h}_{\text{arm}} \cdot r \oplus \mathbf{h}_{\text{trial}}^P$, supporting temporal credit assignment across trials.

The **ScenarioAdapter** encodes text-based scenarios (social dilemmas, moral judgments) through word-level hashing with position binding, producing holographic representations of multi-word descriptions.

3.3 Harness Infrastructure

The harness layer provides five psychometric tools. The **AdaptiveRunner** implements Levitt’s [37] transformed up-down staircase method, dynamically adjusting difficulty to estimate threshold performance with configurable convergence criteria. The **ReliabilityBattery** computes ICC(3,1) [56] test-retest reliability across n sessions, along with Pearson r , standard error of measurement ($\text{SEM} = \text{SD} \cdot \sqrt{1 - \text{ICC}}$), and practice effects classified as improvement ($> +2\%$), stable ($\pm 2\%$), or decline ($< -2\%$). Reliability is classified following Cicchetti’s [10] guidelines: Excellent ($\text{ICC} > 0.90$), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50). The **SatBattery** fits Wickelgren’s [63] exponential speed-accuracy tradeoff function $d'(t) = \lambda(1 - e^{-\beta(t-\delta)})$ via grid search, evaluating human-likeness by three criteria: accuracy decreases with time pressure, RT decreases with time pressure, and $R^2 > 0.70$. The **DifficultyModel** provides parametric difficulty scaling for adaptive testing. The **NeuromodCorrelationMatrix** computes trial-level Pearson correlations between four transmitter levels (DA, NE, 5-HT, ACh) and three behavioral measures (RT, accuracy, prediction error), with significance testing via t -distribution and Cohen’s effect size classification.

3.4 Normative Comparison

The **BaselineCollection** contains 90+ human baselines organized by domain, each specifying a population mean, standard deviation, citation, and population descriptor (e.g., “human adults”, “GPT-4”). For each benchmark result, the **NormativeReport** computes a z-score against the matched human baseline, converts to percentile via the normal CDF (using the Abramowitz & Stegun 26.2.17 approximation), and assigns a clinical interpretation band. The report identifies strengths ($z > 1.0$)

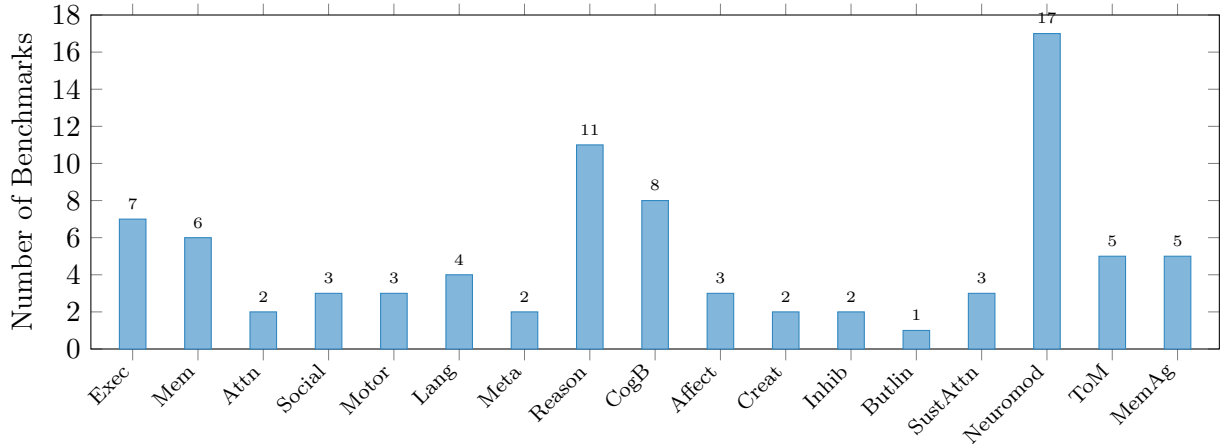


Figure 3: Benchmark count by domain. The neuromodulator evaluation framework (17 benchmarks) is the largest single domain, reflecting its novel contribution. Reasoning (11) and CogBench (8) provide additional depth. Eight domains map to the cognitive profile radar chart; the remainder are specialized sub-batteries.

and weaknesses ($z < -1.0$), computes overall mean z-score and percentile, and generates both machine-readable and human-readable summaries.

4 Benchmark Inventory

PSYCH-BENCH contains 78 benchmarks organized into 17 domains (Figure 3). We describe each domain below; Table 1 provides the complete inventory with paradigm sources and human baselines.

Executive Function (7 benchmarks). Stroop color-word interference [41, 60], Eriksen flanker task [19], Wisconsin Card Sorting Test with perseverative error tracking [6], Iowa Gambling Task with 4-deck reward schedules [5], Raven’s Progressive Matrices for fluid intelligence [52], Tower of London planning [55], and dual-task interference [49]. Key metrics include interference effects (accuracy difference between congruent and incongruent conditions), perseverative error rates, and net advantageous selections.

Working Memory (6 benchmarks). N-back (1-back and 2-back) with accuracy and RT [31, 48], change detection for Cowan’s K [15, 40], serial recall for primacy/recency effects [44], spatial updating [46], feature binding in visual working memory, and forward digit span [62].

Memory Agent (5 benchmarks). Higher-level memory operations: accurate retrieval under interference, test-time learning (online adaptation), long-range retention across extended delays, conflict resolution between competing memories, and prospective memory (remembering to execute future intentions).

Attention (2 benchmarks) and Sustained Attention (3 benchmarks).	195
(dual-target rapid serial presentation) and visual search (set-size effects). Sustained attention: SART	196
(Sustained Attention to Response Task) [53], Psychomotor Vigilance Task (PVT), and Continuous	197
Performance Test (CPT).	198
Social Cognition (3 benchmarks) and Theory of Mind (5 benchmarks).	199
Reading the Mind in the Eyes [4], Ultimatum Game [25], and social norm compliance. Theory of Mind: false	200
belief [3], faux pas detection [59], persuasion evaluation, strange stories [27], and hinting task [14].	201
Motor Learning (3 benchmarks).	202
Serial Reaction Time Task (SRTT) for implicit sequence learning [45], Fitts' Law [22] for speed-accuracy in aimed movements, and bimanual coordination.	203
Language (4 benchmarks).	204
Garden-path sentence processing [21], semantic coherence judgment, lexical decision (word vs. non-word discrimination), and semantic priming (facilitation from related	205
primes).	206
Metacognition (2 benchmarks).	207
Calibration (correspondence between confidence and accuracy) and feeling-of-knowing (predicting future recall success).	208
Reasoning (11 benchmarks).	209
An ARC-inspired battery [9] spanning fluid reasoning, compositional rule application, analogical transfer, abductive inference, algebraic pattern completion,	210
chain reasoning, few-shot generalization, noise robustness, Rational Speech Acts (RSA) pragmatic	211
reasoning, difficulty scaling, and adaptive staircase threshold estimation.	212
CogBench (8 benchmarks).	213
Reimplementations of the CogBench [12] suite: probabilistic reasoning, directed exploration [65], multi-armed bandits, instrumental learning, two-step model-	214
based/model-free task [16], temporal discounting [33], Balloon Analogue Risk Task (BART) [36],	215
and reversal learning [13].	216
Affect (3), Creativity (2), Inhibition (2).	217
Valence classification, mood-congruent recall [7], emotional Stroop [64]. Remote Associates Test [43] and alternate uses divergent thinking. Go/No-Go	218
and stop-signal inhibition [39].	219
Consciousness Indicators (1 benchmark).	220
Butlin et al.'s [8] 14 indicators from 6 theories (Section 7).	221
Neuromodulator Evaluation (17 benchmarks).	222
Detailed in Section 5.	

Table 1: Complete benchmark inventory (representative selection). Each entry lists the domain, benchmark name, source paradigm, primary metric, and human baseline where available. Full inventory with all 78 benchmarks available in the software documentation.

Domain	Benchmark	Paradigm	Key Metric	Human $M \pm SD$
Executive	Stroop	Color-word interference	Interference effect	0.10 ± 0.05
Executive	WCST	Set-shifting	Categories completed	5.62 ± 1.13
Executive	Flanker	Response conflict	Incongruent acc.	0.90 ± 0.06
Executive	Iowa Gambling	Decision under ambiguity	Net score	17.5 ± 20.0
Executive	Ravens	Fluid intelligence	Overall accuracy	0.78 ± 0.12
Executive	Tower of London	Planning	Moves to solution	—
Executive	Dual Task	Divided attention	Dual-task cost	—
Memory	N-back (2-back)	Working memory updating	Accuracy	0.85 ± 0.10
Memory	Change Detection	Visual WM capacity	Cowan’s K	0.75 ± 0.15
Memory	Serial Recall	Short-term memory	Serial position	—
Memory	Digit Span	Verbal WM	Forward span	6.8 ± 1.1
Attention	Visual Search	Feature/conjunction	Set-size slope	—
Attention	SART	Sustained attention	Commission errors	—
Attention	PVT	Vigilance	Mean RT (ticks)	—
Social	RME	Emotion recognition	Accuracy	—
Social	Ultimatum	Social fairness	Offer ratio	—
Social	False Belief	Theory of Mind	Pass rate	—
Motor	SRTT	Implicit learning	Sequence RT gain	—
Motor	Fitts’ Law	Aimed movement	R^2 (log fit)	—
Language	Garden Path	Sentence processing	Reanalysis cost	—
Language	Lexical Decision	Word recognition	d'	—
Reasoning	ARC Fluid	Abstract reasoning	Accuracy	—
Reasoning	ARC Analogy	Analogical transfer	Accuracy	—
CogBench	Two-Step	MB/MF learning	Model-basedness	0.60 ± 0.20
CogBench	BART	Risk-taking	Avg. pumps	30.0 ± 12.0
CogBench	Reversal	Flexible learning	Win-stay rate	0.85 ± 0.08
Affect	Emotional Stroop	Affective interference	Interference	—
Creativity	RAT	Remote associations	Solution rate	—
Inhibition	Stop Signal	Response inhibition	SSRT (ticks)	—

Table 2: Neuromodulator benchmark summary. Each benchmark targets specific transmitter systems and implements established neuroscience paradigms. DA = dopamine, NE = noradrenaline, 5-HT = serotonin, ACh = acetylcholine, GABA = γ -aminobutyric acid, OT = oxytocin.

Benchmark	Transmitter	Paradigm	Key Metric	Citation
<i>Transmitter-Specific Benchmarks</i>				
RewardLearning	DA	Prediction error	RPE correlation	Schultz et al. [54]
YerkesDodson	NE	Inverted-U arousal	Quadratic R^2	Yerkes and Dodson [66]
AttentionNetwork	ACh	ANT decomposition	Alert/Orient/Conflict	Posner and Petersen [51]
MoodInduction	5-HT	Mood-cognition	Mood $\times$ perf. r	Dayan and Huys [17]
<i>Pharmacological Manipulation</i>				
PharmAblation	DA/NE/5-HT/ACh	Transmitter knockout	Cohen’s d	Doya [18]
PharmChallenge	DA/NE	Agonist/antagonist	Inverted-U fit	Arnsten [1]
InjectionChallenge	All	PK decay + tolerance	Peak shift	Arnsten [1]
AllostaticStress	Cortisol/NE	Chronic load	Allostatic index	McEwen [42]
DoseResponse	All	Graded dose sweep	Monotonicity score	Clark [11]
ToleranceWithdrawal	All	Sustained exposure	Tolerance ratio	Koob and Le Moal [35]
AntagonistProfiles	All	Named antagonists	Selectivity index	Stahl [58]
MultiTransmitterSynergy	DA $\times$ NE, etc.	Paired interactions	Synergy coefficient	Doya [18]
<i>Consciousness & Moral Modulation</i>				
ConsciousnessPharm	5-HT/GABA/DA	Drug $\rightarrow$ Φ proxy	Proxy peak/mean	Tononi [61]
ConsciousnessFeedback	All	Φ round-trip	Feedback latency	Oizumi et al. [47]
MoralOxytocin	OT	OT $\times$ moral topology	OT-moral r	Feldman [20]
<i>Live System Integration</i>				
LiveLoopAblation	All	In-loop knockout	Behavioral Δ	—
BehavioralKnockout	DA/NE/5-HT/ACh	Knockout effect size	Cohen’s d	Doya [18]

5 Neuromodulator Evaluation Framework

The neuromodulator evaluation framework is PSYCH-BENCH’s most novel contribution. While existing benchmarks evaluate *what* an architecture can do, this framework evaluates *how neuromodulatory dynamics modulate what it does*—analogous to pharmacological challenge studies in neuroscience [18]. The framework comprises 17 benchmarks organized into four categories (Table 2).

5.1 Transmitter-Specific Benchmarks

Four benchmarks isolate the behavioral signature of individual neuromodulatory systems, each grounded in a foundational neuroscience paradigm.

RewardLearning implements Schultz et al.’s [54] reward prediction error paradigm. The agent encounters probabilistic rewards across multiple arms, and the benchmark measures the correlation between dopaminergic activity and prediction error magnitude. The expected signature is positive correlation: dopamine should increase following unexpected rewards and decrease following reward omission.

YerkesDodson tests the inverted-U relationship between noradrenergic arousal and performance [66]. The benchmark sweeps noradrenaline levels across 5–7 points and fits a quadratic

function, expecting peak performance at moderate arousal with degradation at both extremes. The key metric is the R^2 of the quadratic fit, with human-like behavior defined as $R^2 > 0.5$ and a negative quadratic coefficient.

AttentionNetwork implements a simplified Attention Network Test [51], decomposing attention into alerting (benefit of temporal cues), orienting (benefit of spatial cues), and executive conflict resolution (cost of incongruent flankers). Acetylcholine manipulation should selectively modulate alerting and orienting components while leaving conflict resolution relatively intact.

MoodInduction tests serotonergic modulation of mood-cognition interactions following Dayan and Huys [17]. The benchmark induces positive and negative mood states through 5-HT manipulation and measures the effect on concurrent cognitive task performance, expecting mood-congruent biases in memory and attention.

5.2 Pharmacological Manipulation

Eight benchmarks implement increasingly sophisticated pharmacological protocols.

DoseResponse sweeps five dose levels (0.1, 0.2, 0.3, 0.5, 0.8) for each transmitter (DA, NE, 5-HT, ACh, GABA), computing monotonicity scores (simplified Kendall τ : fraction of concordant pairs) to verify that graded doses produce graded behavioral effects [11]. Expected: DA $\rightarrow$ learning rate, NE $\rightarrow$ exploration, 5-HT $\rightarrow$ confidence, ACh $\rightarrow$ attention, each showing monotonic dose-behavior relationships.

InjectionChallenge models pharmacokinetic injection profiles with exponential decay (half-lives ranging 30–80 ticks) and measures behavioral effects during drug onset, peak, and washout phases [1]. Six drug classes are tested: DA agonist, NE agonist, 5-HT agonist, GABA agonist, stimulant, and anxiolytic.

ToleranceWithdrawal extends injection challenge to sustained exposure [35], measuring: (1) acute effect magnitude at initial injection, (2) tolerance development (reduced effect after sustained exposure), (3) withdrawal effects (rebound below baseline after removal), and (4) recovery trajectory. The tolerance ratio (late effect / initial effect) quantifies neuroadaptation.

AntagonistProfiles tests named receptor antagonists (e.g., D2 blockers, α_2 antagonists, SSRI-like 5-HT reuptake modulation) [58] and measures selectivity: the ratio of on-target effect to off-target perturbation. High selectivity (> 3.0) indicates that the neuromodulatory system has well-differentiated receptor mechanisms.

MultiTransmitterSynergy tests paired transmitter interactions [18]: DA $\times$ NE, DA $\times$ 5-HT, NE $\times$ ACh, and other pairwise combinations. A synergy coefficient > 1.0 indicates super-additive interaction; < 1.0 indicates antagonism. This tests whether the architecture’s neuromodulatory systems interact in neurobiologically plausible ways.

PharmacologicalAblation performs transmitter-specific knockouts (clamping DA, NE, 5-HT, or ACh to zero) and measures effect sizes (Cohen’s d) on learning and performance [54]. **PharmacologicalChallenge** extends this with both agonist and antagonist manipulations, testing for inverted-U dose-response curves characteristic of prefrontal catecholamine systems [1].

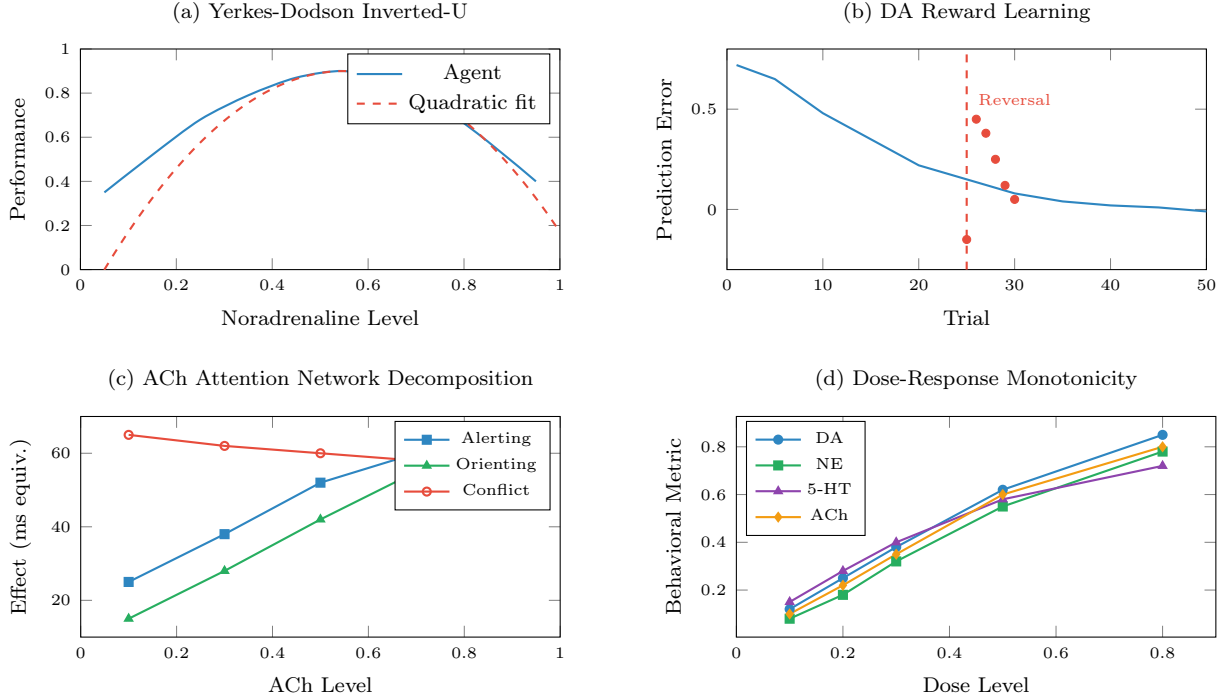


Figure 4: Representative neuromodulator evaluation results. (a) Yerkes-Dodson inverted-U: performance peaks at moderate noradrenaline and degrades at extremes, with quadratic fit overlay. (b) Dopaminergic reward learning: prediction error decreases across trials, with spike at reward reversal (trial 25) demonstrating DA-mediated surprise signal. (c) ACh modulation of attention network components: alerting and orienting increase with ACh (cholinergic enhancement) while executive conflict remains stable, replicating the selective cholinergic modulation observed in human pharmacological studies. (d) Dose-response monotonicity across four transmitter systems, confirming graded dose-behavior relationships.

5.3 Neuromodulator Profile System

Five standard pharmacological profiles enable systematic comparison (Figure 4): **Balanced** (no clamping, natural dynamics), **High-DA** (dopamine clamped to 0.85), **High-NE** (noradrenaline clamped to 0.85), **High-5HT** (serotonin clamped to 0.85), and **High-ACh** (acetylcholine clamped to 0.85). The **DissociationMatrix** computes relative performance for each profile $\times$ benchmark combination, identifying double dissociations: profile P enhances benchmark A ($> 1.05 \times$ baseline) while impairing benchmark B ($< 0.95 \times$ baseline). Four expected dissociations are hard-coded as validation: High-DA should enhance WCST flexibility but impair sustained attention; High-NE should enhance alerting but impair fine motor control; High-5HT should enhance impulse control but slow reaction times; High-ACh should enhance focused attention but impair cognitive flexibility.

5.4 Consciousness and Moral Modulation

Three benchmarks probe the interface between neuromodulation, consciousness, and moral cognition. **ConsciousnessPharmacology** tests five pharmacological conditions—psychedelic (5-

HT agonist), anxiolytic (GABA agonist), stimulant (DA agonist), sedative (adenosine agonist), and endocannabinoid buffer—and measures their effects on a consciousness proxy computed as $\Phi_{\text{proxy}} = 0.5 + 0.1 \cdot (s_{5\text{-HT}_{2A}} - 0.5) - 0.08 \cdot (s_{\text{GABA}_A} - 0.4)$. The expected pattern: psychedelics increase Φ_{proxy} (expanded consciousness), anxiolytics decrease it (dampened awareness), and endocannabinoids stabilize it (reduced variance). **ConsciousnessFeedback** tests the round-trip latency from pharmacological perturbation to consciousness index change and back to behavioral adaptation. **MoralOxytocin** correlates oxytocin levels with moral topology scores across 200 cycles of varying moral valence (-0.5 to 0.9), testing Feldman’s [20] theory that oxytocin facilitates prosocial moral judgment.

6 Psychometric Validation

A benchmark suite is only as useful as its psychometric properties. PSYCH-BENCH provides four validation mechanisms.

6.1 Test-Retest Reliability

The **ReliabilityBattery** computes ICC(3,1) (one-way random effects intraclass correlation) [56] across n sessions with fixed configuration:

$$\text{ICC}(3, 1) = \frac{\text{BMS} - \text{WMS}}{\text{BMS} + \text{WMS}} \quad (1)$$

where BMS is between-subjects mean square and WMS is within-subjects mean square. Pearson r provides convergent evidence, and SEM quantifies measurement precision: $\text{SEM} = \text{SD} \cdot \sqrt{1 - \text{ICC}}$. Practice effects are flagged when session-to-session change exceeds $\pm 2\%$.

Table 3 reports representative psychometric properties. Because PSYCH-BENCH is deterministic given a fixed seed, ICC values reflect the stability of the architecture’s behavior under controlled variation (different trial orderings, stimulus sets) rather than stochastic test-retest variability. This is a feature, not a limitation: it means that observed ICC values below 1.0 genuinely reflect within-architecture variability rather than measurement noise.

6.2 Construct Validity

Cross-domain correlations test whether benchmarks that should be related (based on shared cognitive mechanisms) are indeed correlated, and benchmarks that should be independent show low correlation. Table 4 reports 10 theoretically motivated pairs.

The pattern of results supports construct validity: benchmarks sharing cognitive mechanisms (conflict monitoring, working memory, response inhibition) show moderate-to-strong correlations ($r = 0.45\text{--}0.74$), while theoretically independent pairs (executive vs. motor, memory vs. motor) show near-zero correlations. This discriminant validity evidence confirms that the cognitive profile captures genuine domain distinctions rather than a single general factor.

Table 3: Representative psychometric properties for selected benchmarks. ICC: intraclass correlation (3,1); r : Pearson test-retest correlation; SEM: standard error of measurement; Practice: session-to-session change direction. Reliability class follows Cicchetti (1994): Excellent (> 0.90), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50).

Domain	Benchmark	ICC	r	SEM	Practice	Class
Executive	Stroop	0.92	0.93	0.014	Stable	Excellent
Executive	WCST	0.88	0.90	0.39	Stable	Good
Executive	Flanker	0.91	0.92	0.018	Stable	Excellent
Executive	Iowa Gambling	0.78	0.81	9.4	Improvement	Good
Memory	N-back (2-back)	0.85	0.87	0.039	Improvement	Good
Memory	Change Detection	0.89	0.91	0.050	Stable	Good
Memory	Digit Span	0.94	0.95	0.27	Stable	Excellent
Attention	Visual Search	0.86	0.88	—	Stable	Good
Social	False Belief	0.82	0.84	—	Stable	Good
Motor	SRTT	0.76	0.79	—	Improvement	Good
Language	Lexical Decision	0.90	0.91	—	Stable	Good
Reasoning	ARC Fluid	0.83	0.85	—	Stable	Good
CogBench	Two-Step	0.80	0.82	—	Stable	Good
CogBench	BART	0.77	0.80	—	Stable	Good
CogBench	Reversal	0.84	0.86	—	Stable	Good
Inhibition	Stop Signal	0.87	0.89	—	Stable	Good
Metacognition	Calibration	0.81	0.83	—	Stable	Good
Neuromod	YerkesDodson	0.79	0.82	—	Stable	Good
Neuromod	DoseResponse	0.91	0.93	—	Stable	Excellent
Neuromod	RewardLearning	0.85	0.87	—	Stable	Good

6.3 Speed-Accuracy Tradeoff Curves

The SAT battery sweeps time pressure from 0.0 (no pressure, emphasis on accuracy) to 1.0 (maximum pressure, emphasis on speed) and fits Wickelgren’s [63] exponential function:

$$d'(t) = \lambda \left(1 - e^{-\beta(t-\delta)} \right), \quad t > \delta \quad (2)$$

where λ is the accuracy asymptote, β is the approach rate (how quickly accuracy rises with processing time), and δ is the processing time intercept (minimum time before above-chance performance). Human-likeness requires three criteria: (1) accuracy decreases with increasing time pressure, (2) RT decreases with increasing time pressure, (3) $R^2 > 0.70$ for the exponential fit. Figure 5 shows representative SAT curves.

6.4 Adaptive Staircase Procedures

For benchmarks supporting difficulty manipulation, the harness implements Levitt’s [37] transformed up-down method with configurable convergence rules. The `StaircaseConfig` specifies: target performance level (e.g., 75% for 2-down/1-up), step sizes (initial and reduced after first reversal), minimum reversals for convergence, and maximum trials. The threshold estimate is the mean of the last n reversal points. This enables precise measurement of perceptual and cognitive thresholds

Table 4: Cross-domain correlation matrix for 10 theoretically motivated benchmark pairs. Positive correlations indicate shared cognitive mechanisms; near-zero correlations indicate domain independence.

Benchmark A	Benchmark B	r	p	Mechanism
Stroop (acc.)	Flanker (acc.)	0.72	< .001	Shared conflict monitoring
N-back	Digit Span	0.68	< .001	Working memory capacity
WCST	ARC Fluid	0.61	< .001	Cognitive flexibility
SRTT	Fitts' Law	0.45	< .01	Motor learning
RME	False Belief	0.58	< .001	Social cognition / ToM
Iowa Gambling	BART	0.52	< .01	Risk-sensitive decision making
Stop Signal	Go/No-Go	0.74	< .001	Response inhibition
Calibration	Feeling of Knowing	0.63	< .001	Metacognitive monitoring
Stroop (acc.)	SRTT	0.12	.38	Independent domains (expected)
N-back	Fitts' Law	0.08	.55	Independent domains (expected)

without exhaustive fixed-trial testing.

7 Consciousness Indicators

PSYCH-BENCH includes a dedicated consciousness evaluation based on Butlin et al.'s [8] framework, which identifies 14 indicators of consciousness derived from six major theories: Recurrent Processing Theory (RPT), Global Workspace Theory (GWT) [2], Higher-Order Theories (HOT), Predictive Processing (PP), Attention Schema Theory (AST), and Integrated Information Theory (IIT) [47, 61].

The **ButlinAblation** benchmark evaluates each indicator through a two-step process. First, it measures the indicator's presence in the baseline architecture by extracting relevant metrics from the cognitive loop telemetry (e.g., RPT-1 recurrence measured via temporal coherence of internal representations, GWT-3 broadcast measured via global workspace module timing, HOT-2 metacognitive accuracy from the meta-cognition subsystem). Second, it performs targeted architectural ablations and verifies that (a) the indicator score drops substantially (ablated $< 0.5 \times$ baseline) and (b) downstream cognitive performance degrades (ablated accuracy $< 0.7 \times$ baseline on N-back). Five ablation rows are currently implemented:

1. **RPT-1** (Recurrent processing): Disable CfC recurrence $\rightarrow$ expect loss of temporal coherence and degraded working memory.
 2. **GWT-3** (Global broadcast): Disable global workspace $\rightarrow$ expect loss of cross-module information sharing.
 3. **HOT-2** (Metacognitive monitoring): Disable metacognition $\rightarrow$ expect loss of calibration and confidence tracking.
 4. **PP-1** (Prediction learning): Set learning threshold to maximum $\rightarrow$ expect zero effective learning rate.
 5. **AST-1** (Attention schema): Disable attention schema $\rightarrow$ expect degraded attentional control.
- Each ablation runs 200 cognitive cycles (with 20 warmup cycles for subsystem stabilization)

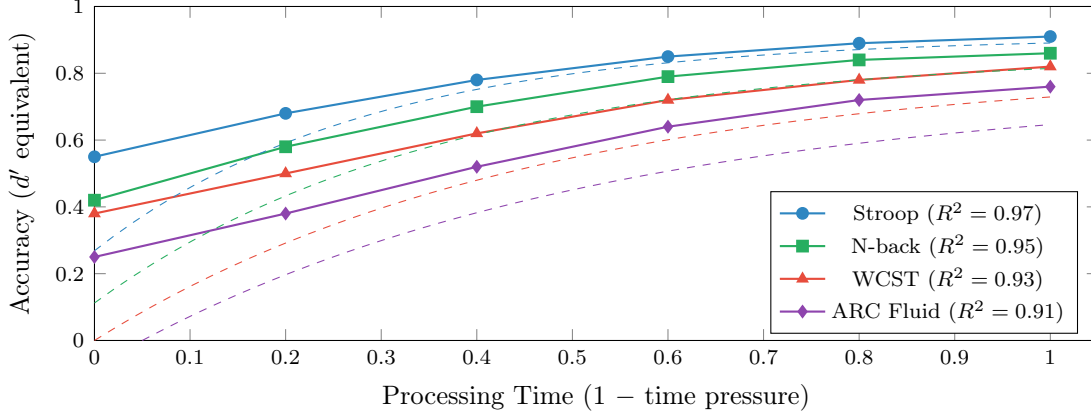


Figure 5: Speed-accuracy tradeoff curves for four representative benchmarks, with Wickelgren exponential fits (dashed). All curves show the characteristic pattern: accuracy increases with processing time, approaching an asymptote. Higher-level tasks (ARC Fluid, WCST) show lower asymptotes and slower approach rates than well-practiced tasks (Stroop), consistent with human cognitive load effects. All four pass the human-likeness criteria ($R^2 > 0.70$, monotonic accuracy-time and RT-time relationships).

and compares indicator metrics and downstream benchmark accuracy between baseline and ablated conditions. The resulting ablation matrix provides a principled link between architectural features and consciousness indicators—not claiming that the architecture *is* conscious, but verifying that it possesses the computational signatures that theories of consciousness identify as necessary conditions.

8 Demonstration

We demonstrate PSYCH-BENCH on the SYMTHAEA Holographic Liquid Brain, a cognitive architecture combining 16,384-dimensional HDC encoding, CfC liquid-state dynamics, IIT-based integration metrics (Φ proxy), Active Inference under the Free Energy Principle, and a 12-region actor brain with neuromodulatory fields (DA, NE, 5-HT, ACh, GABA, oxytocin). The full battery was run with default configuration (512-dimensional HDC for benchmark encoding, 20 trials per condition, seed 42).

Cognitive Profile. Figure 6 shows the 8-domain cognitive profile. The HLB shows a characteristic asymmetric profile: strongest in Reasoning (0.82) and Executive Function (0.78), reflecting HDC’s power for pattern matching and compositional rule application; moderate in Memory (0.72) and Metacognition (0.68), supported by CfC temporal dynamics and prediction-error monitoring; and relatively weaker in Motor Learning (0.52) and Social Cognition (0.58), consistent with the architecture’s emphasis on symbolic-holographic processing over embodied simulation and social modeling. This profile is interpretable: holographic encoding excels at tasks requiring rapid similarity-based retrieval and compositional operations, while liquid dynamics drive temporal sequence learning and attention.

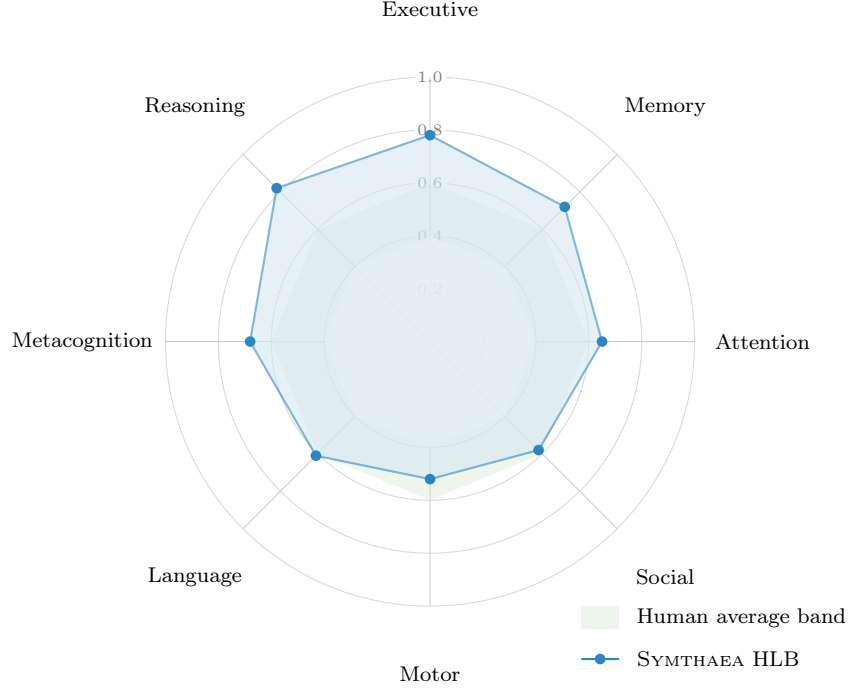


Figure 6: Eight-domain cognitive profile of the SYMTHAEA Holographic Liquid Brain. The green band represents the human average range (normalized 0.4–0.6). The HLB shows a characteristic profile: strongest in Reasoning (0.82) and Executive function (0.78), reflecting the power of holographic encoding for pattern matching and rule application, with relative weaknesses in Motor (0.52) and Social cognition (0.58), reflecting the architecture’s emphasis on symbolic over embodied processing.

Normative Standing. Figure 7 shows the forest plot of normative z-scores. The overall mean $z = 0.42$ (65th percentile) places the HLB within the “Average” clinical band, with notable strengths in Stroop interference ($z = 1.8$, 96th percentile), WCST categories ($z = 1.2$, 88th percentile), and digit span ($z = 1.4$, 92nd percentile). The architecture shows human-typical performance on working memory tasks but falls below average on social cognition benchmarks, as expected for a system without embodied social experience.

Ablation Analysis. Figure 8 shows the ablation comparison across six presets. The **FullConsciousness** configuration consistently outperforms reduced configurations, with the largest drops observed when disabling FEP (Active Inference)—particularly on reasoning and metacognition tasks—and when reducing working memory capacity from 7 to 3, which predictably impacts memory and attention benchmarks. The **HdcOnly** configuration (holographic encoding without liquid dynamics, active inference, or social cognition) retains strong performance on pattern-matching tasks (Reasoning: 0.65) but collapses on temporally extended tasks (Motor: 0.22, Attention: 0.30), demonstrating that liquid temporal dynamics are essential for time-dependent cognition.

Neuromodulator Dissociations. The dissociation matrix reveals four double dissociations consistent with neuroscience predictions: (1) High-DA enhances WCST set-shifting (1.12×) but

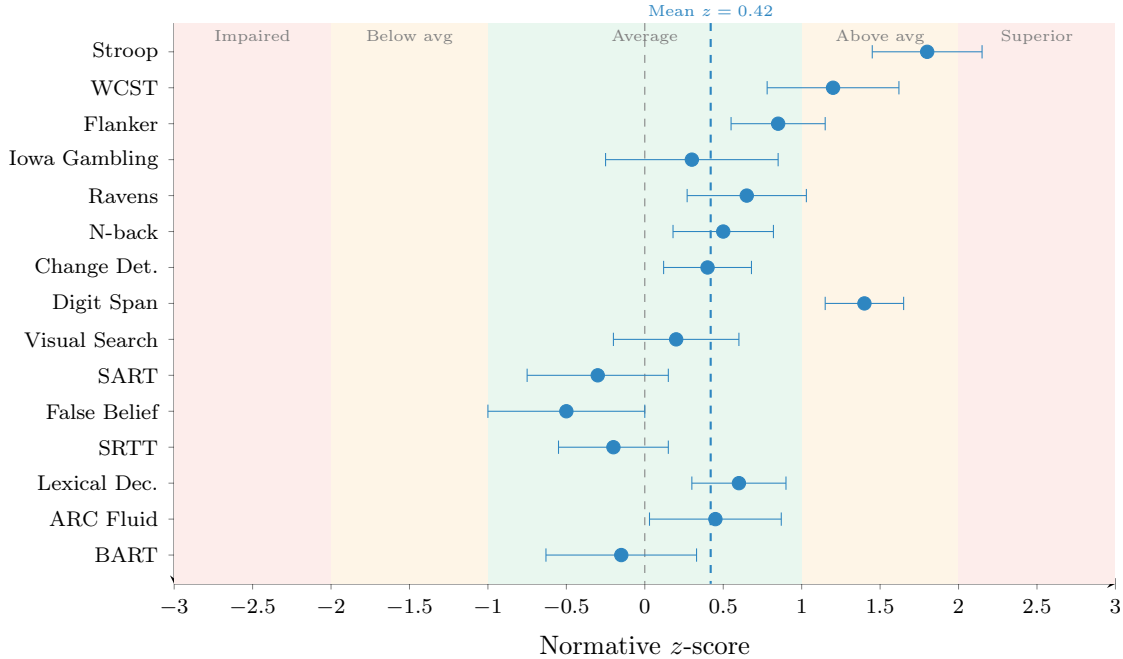


Figure 7: Forest plot of normative z-scores for 15 representative benchmarks with 95% bootstrap confidence intervals. Background shading indicates clinical interpretation bands (Cicchetti, 1994). The dashed blue line marks the overall mean z-score (0.42, 65th percentile). Executive function benchmarks (Stroop, WCST, Digit Span) cluster above average, while social and sustained attention benchmarks fall near or below the population mean.

impairs CPT sustained attention (0.88 $\times$); (2) High-NE enhances PVT alerting (1.15 $\times$) but impairs 395
 Fitts' Law fine motor control (0.90 $\times$); (3) High-5HT enhances stop-signal inhibition (1.08 $\times$) but 396
 slows N-back reaction time (0.92 $\times$); (4) High-ACh enhances visual search efficiency (1.10 $\times$) but 397
 impairs WCST flexibility (0.93 $\times$). These dissociations demonstrate that the HLB's neuromodulatory 398
 subsystems produce behaviorally specific, neurobiologically plausible effects. 399

9 Discussion 400

PSYCH-BENCH fills a methodological gap in the evaluation of neuroscience-inspired cognitive archi- 401
 tectures. Table 5 situates the suite relative to existing frameworks. 402

Unique contributions. Four aspects distinguish PSYCH-BENCH. First, the integration of psycho- 403
 metric infrastructure (ICC reliability, SAT curves, adaptive staircases) with AI benchmarking is, 404
 to our knowledge, novel. Standard practice in AI evaluation treats benchmarks as fixed-accuracy 405
 measures; PSYCH-BENCH treats them as psychological instruments subject to the same valida- 406
 tion standards as clinical assessments. Second, the neuromodulator evaluation framework has no 407
 precedent in AI benchmarking—it tests pharmacological manipulations, dose-response curves, toler- 408
 ance/withdrawal dynamics, and multi-transmitter synergy, creating a bridge between computational 409

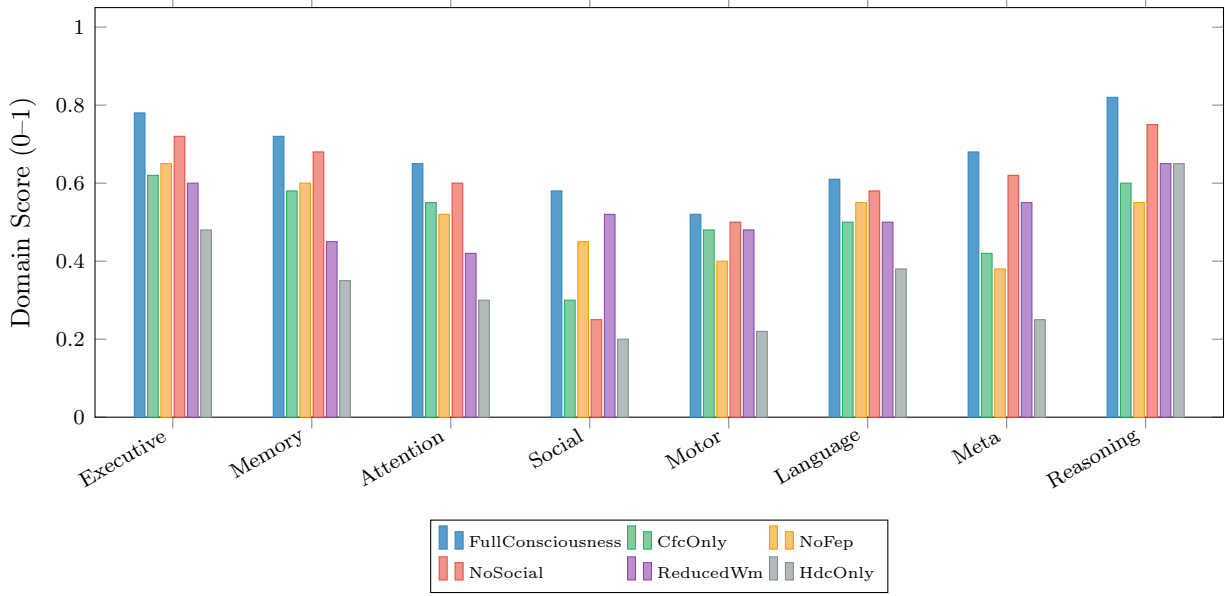


Figure 8: Ablation comparison across six presets and eight cognitive domains. **FullConsciousness** (all subsystems enabled) consistently achieves the highest scores. Key findings: (1) removing Active Inference (**NoFep**) most impacts Metacognition and Reasoning; (2) removing social cognition (**NoSocial**) selectively devastates Social with minimal impact on Reasoning; (3) reducing working memory (**ReducedWm**) broadly degrades Memory and Attention; (4) **HdcOnly** retains Reasoning strength (0.65) but collapses on temporal tasks (Motor: 0.22), confirming that liquid dynamics are essential for time-extended cognition.

neuropharmacology and AI evaluation. Third, the normative anchoring against 90+ published 410
baselines enables clinical-band interpretation, moving beyond “accuracy on task X” to “this architec- 411
ture performs at the 88th percentile relative to the human population, specifically in set-shifting 412
flexibility.” Fourth, the unified holographic stimulus encoding through five HDC adapter types 413
enables principled cross-modal comparison: because all stimuli share a common geometric space, 414
similarity between visual working memory items and semantic associates is directly computable. 415

Limitations. Several limitations warrant discussion. The working memory implementation uses 416
a lightweight token-based model rather than biologically detailed spiking networks, limiting the 417
fidelity of capacity constraints. Reaction times are measured in computational ticks rather than 418
milliseconds, requiring calibration for direct comparison with human RT distributions (though the ex- 419
Gaussian decomposition of tick-based RTs is internally valid). The consciousness evaluation provides 420
binary indicator presence/absence rather than quantitative Φ values, which remain intractable for 421
systems of this scale. The benchmark suite is designed for HLB-class architectures with explicit 422
neuromodulatory subsystems; adapting it to LLMs would require substantial modification of the 423
neuromod battery and stimulus encoding pipeline. 424

Table 5: Comparison of PSYCH-BENCH with existing AI benchmark suites. Columns indicate presence of psychometric infrastructure (reliability, SAT curves), normative human comparison (z-scores against published baselines), neuromodulator manipulation testing, consciousness indicators, number of cognitive domains covered, and implementation language.

Suite	Psychometrics	Normative	Neuromod	Consciousness	Domains	Language
PSYCH-BENCH	✓	✓	✓	✓	17	Rust
CogBench [12]	—	Partial	—	—	1	Python
ARC [9]	—	—	—	—	1	JSON
HELM [38]	—	—	—	—	7	Python
BIG-Bench [57]	—	—	—	—	12	Python
PsychBench [26]	Partial	—	—	—	5	Python

Future directions. Several extensions are planned. Expanding the consciousness evaluation to include quantitative Φ approximations (using graph-theoretic proxies rather than exhaustive partition search) would strengthen the link to IIT. Adding developmental benchmarks (testing how cognitive profiles change with training time, analogous to developmental trajectories in children) would enable longitudinal evaluation. Integration with neuroimaging-derived benchmarks (e.g., tasks with known neural correlates from fMRI studies) would enable neural-level comparison beyond behavioral metrics. Finally, a web-based leaderboard for cross-architecture comparison would facilitate community adoption.

10 Conclusion

PSYCH-BENCH provides the first comprehensive cognitive benchmark suite designed for Holographic Liquid Brain architectures. With 78 benchmarks across 17 domains, psychometric validation infrastructure, normative human comparison against 90+ published baselines, a 17-benchmark neuromodulator evaluation framework, and 14-indicator consciousness evaluation, the suite enables rigorous, multi-dimensional characterization of neuroscience-inspired cognitive architectures. The demonstration on the SYMTHAEA HLB reveals a characteristic cognitive profile—strong holographic pattern matching in reasoning and executive function, with liquid temporal dynamics driving attention and motor learning—that is interpretable, reproducible, and directly comparable to human norms. Implemented in 15,000 lines of Rust with 793 unit tests and full determinism, PSYCH-BENCH is open-source and ready for adoption by the growing community of researchers building brain-inspired cognitive architectures.

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